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Disentangling climate signals from survey effort in new bird distribution records across China

Birds or birders? A discrete-time hazard model of first records, 2002–2024

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Background

A new record is the last link in a chain of three events

New distribution record: the first documented occurrence of a species in a province where it had not been recorded before

1 The bird is there



The species occupies the province; warming can move ranges

Ecological process

2 Someone looks



A birder or a survey detects it; effort grew fast after 2000

Observation process

3 The find is published



The detection is written up and dated in a paper or report

Documentation

New record

Only this link is ecological

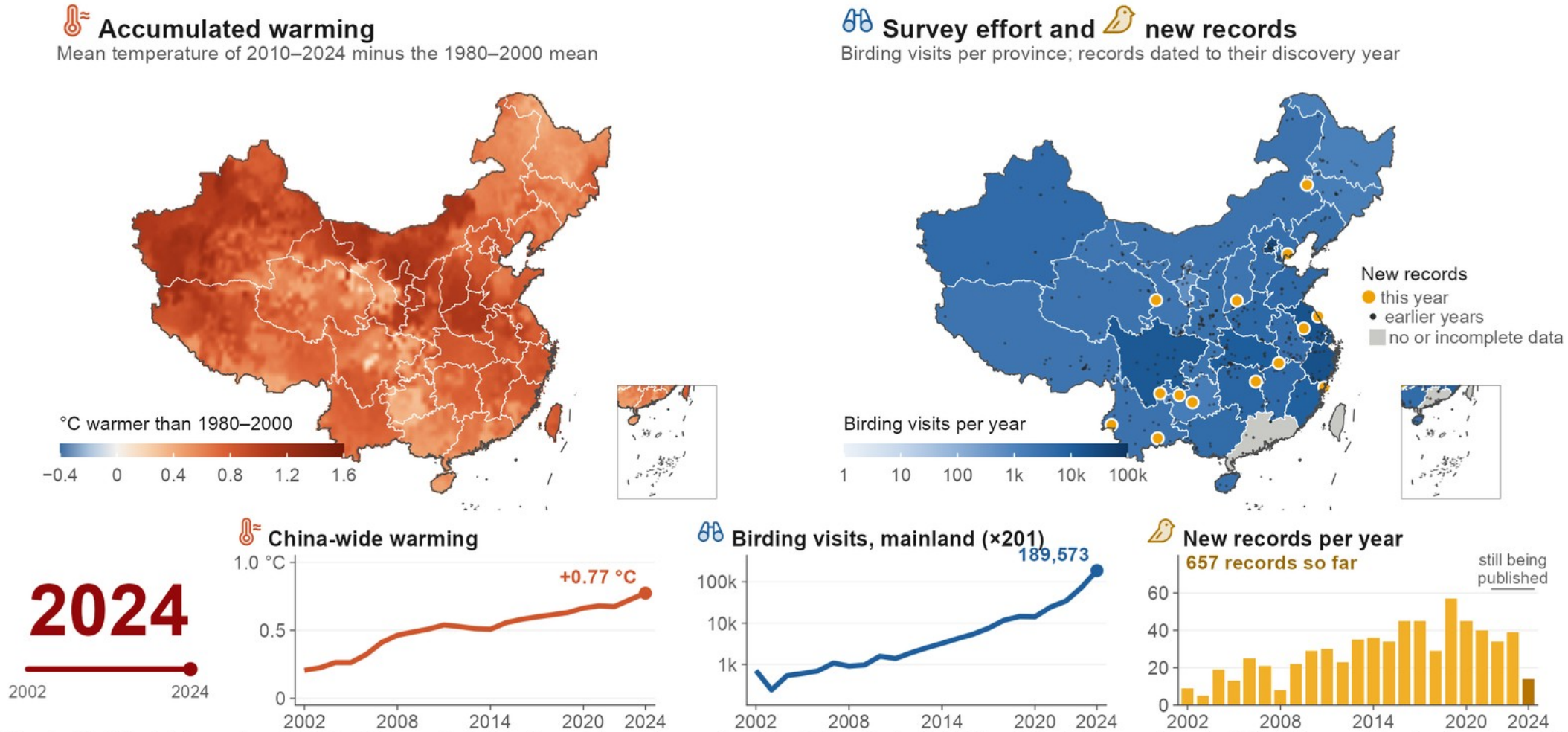
These two leave the same trace in the data

- ☐ Imperfect detection and uneven sampling can create apparent range changes where none occurred
- ☐ In China, provincial new bird records have been read as evidence of poleward shifts

Tingley & Beissinger 2009; Hughes et al. 2021; Xing et al. 2024; Chen et al. 2025

Background

After 2002, warming, birding and new records all rose together



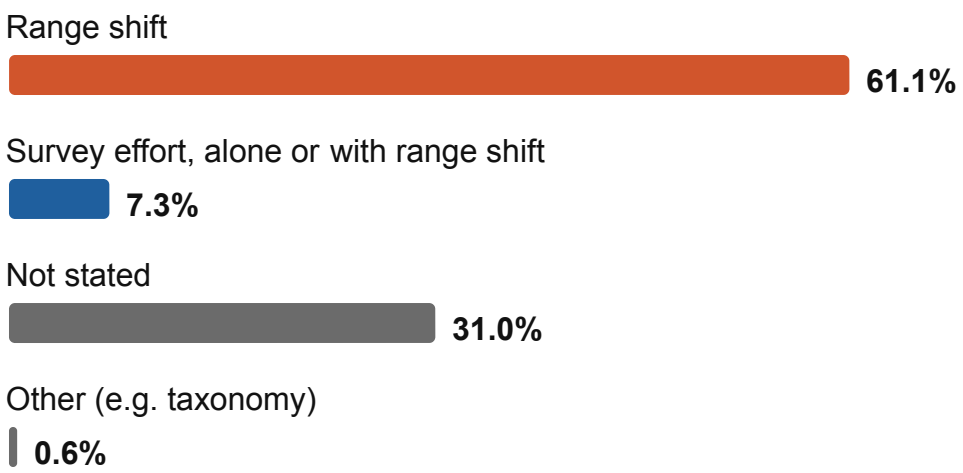
Warming: WorldClim 2.1 downscaling of CRU TS 4.09, the provincial side of the model's warming covariate. Effort: birding visits in eBird/GBIF and China Bird Report. Records: CBNR v2 first records dated to discovery, n = 657. Grey: effort coverage gap; outside the analysis (Taiwan, Hong Kong, Macao); or incomplete download (Guangdong 2024). Base map: GS(2019)1822.

Research gaps

Analyses of new records have not separated ecology from observation

How source papers explain their records

CBNR v2, n = 1,029 provincial first records



Birding visits rose about 200-fold, 2002–04 to 2022–24, yet effort is rarely given as the reason

1 GAP — Confounded drivers

Warming and observation grew together; a model with only one of them credits it with both

2 GAP — Unmodelled observation

Survey pulses, and which species could be recorded where, are rarely in the model

3 GAP — No benchmark, no support check

Warming lacks a null benchmark; projections are rarely checked against the fitted data

Needed: one model that treats each record as the product of ecology and observation

Questions

Four questions, each with a testable expectation

Q1 Does warming raise the rate of new records once observation is modelled?

H1 Yes, beyond the benchmark that shared regional climate alone produces

Q2 How large is the observation process itself?

H2 Province-year survey pulses are the largest source of variation

Q3 Do warming and survey effort interact?

H3 Negatively: where effort is high, species favoured by warming are already recorded

Q4 Are the returns on warming and on effort stable through time?

H4 Returns on effort decline as gaps are filled; returns on warming do not

Approach

From records to a hazard model, in four steps

1 Records

- **CBNR v2** — 1,029 provincial first records
- **Dated to discovery** — 657 records in 2002–2024
- **Current release, v1.1** — 1,076 records · 588 species

2 Risk set

- **Species × province × year** — 175,640 cells, 2002–2024
- **Reachable provinces only** — ≥ 50 suitable SDM cells
- **Exit after first record** — each pair counted once

3 Covariates

- **Warming** — vs species range, 15 yr
- **Survey effort** — birding visits
- **Variability** — annual anomaly
- **Completeness** — offset, recent years

4 Analysis

- **Hazard GLMM** — cloglog link
- **Robustness** — nulls · 12 specs
- **Through time** — time-varying returns
- **Projections** — support checked

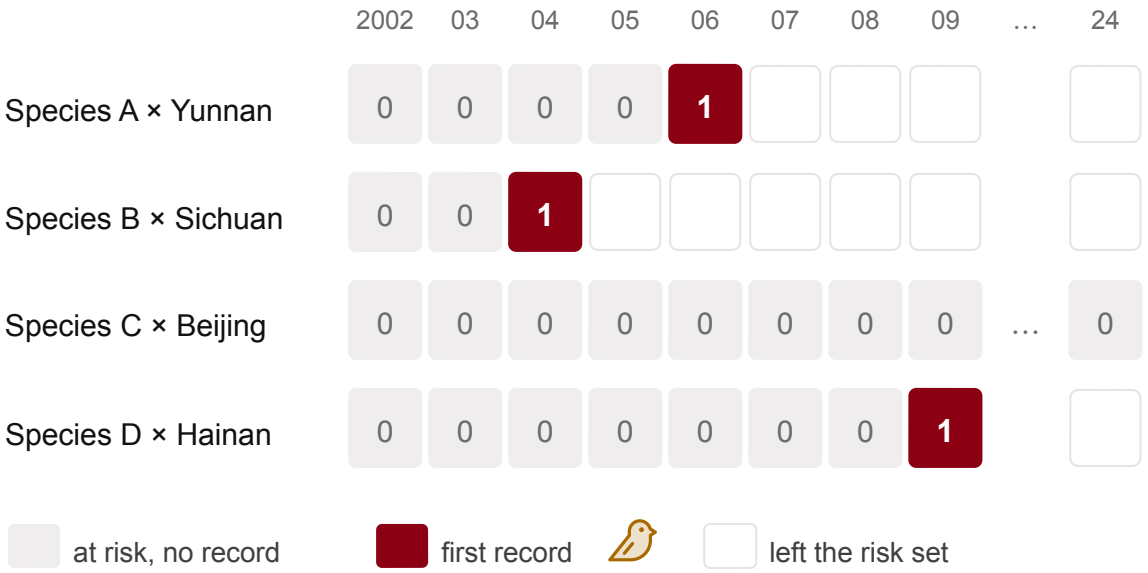
Outputs: contributions · interaction · observation process · returns through time · futures

Approach

A discrete-time hazard model asks: is this the year of the first record?

The risk set

Each pair is at risk every year until its first record



175,640 cells · 649 first records

392 species · 31 provinces · 2002–2024

first record ~ warming × effort + variability
+ offset(log completeness)

+ (1 | species) + (1 | province) + (1 | province × year)

Binomial, cloglog link: exp(β) is a hazard ratio per SD

Accumulated warming

Province minus species-range anomaly, 15-yr; 1 SD = 0.18 °C

Survey effort

Provincial birding visits; 1 SD = 7.9-fold more visits

Province × year effect

Survey campaigns, birding festivals, new reserves

Dated to discovery

Offset corrects the latest, incompletely published years

◆ Finding 1

Warming and survey effort contribute comparably, and they interact

Effort and warming contribute comparably

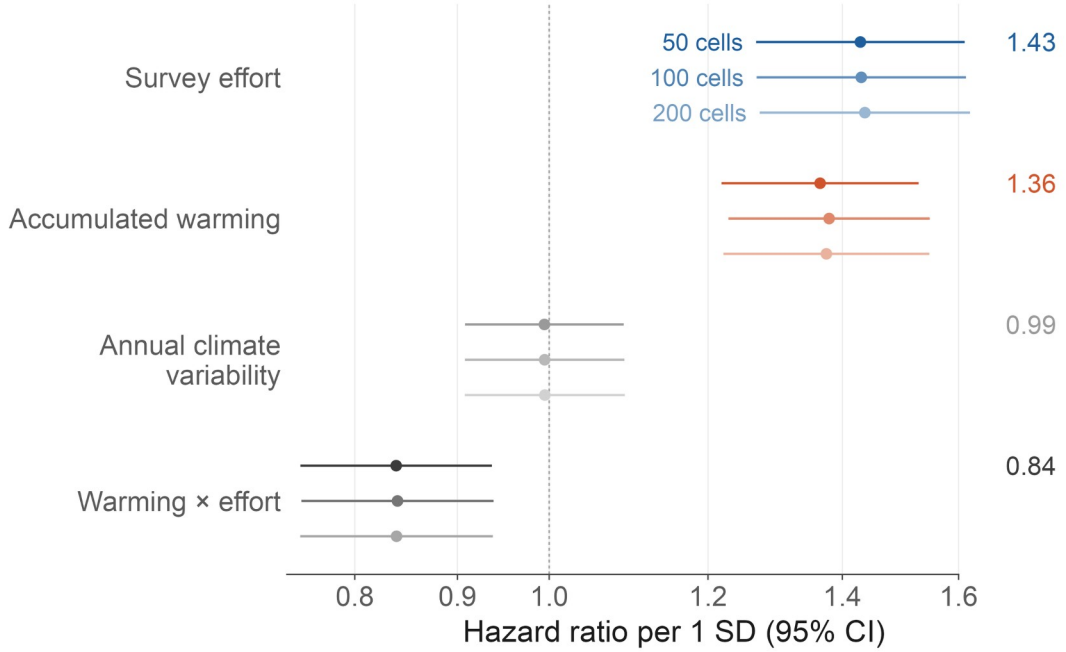


Fig. 1a Hazard ratio per SD, three SDM thresholds

Warming matters most where effort is low

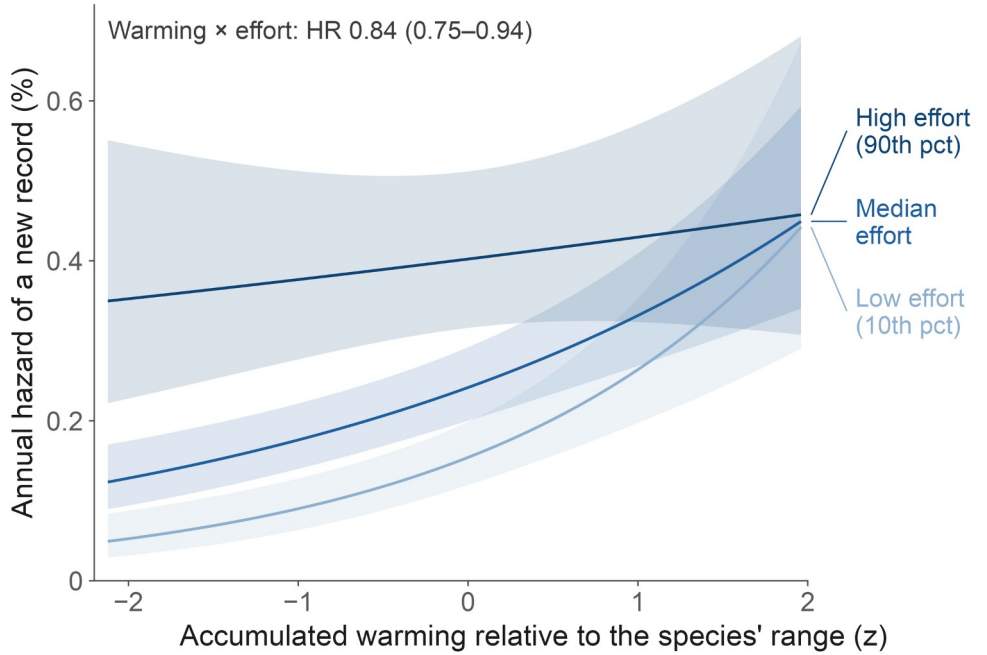


Fig. 1b Effect of warming at three effort levels

Accumulated warming
1.36
HR per SD (1.22–1.53)

Survey effort
1.43
HR per SD (1.27–1.61)

Warming × effort
0.84
steepest where few look

Annual variability
0.99
no effect (0.91–1.09)

Finding 1

The balance holds across four criteria and twelve specifications

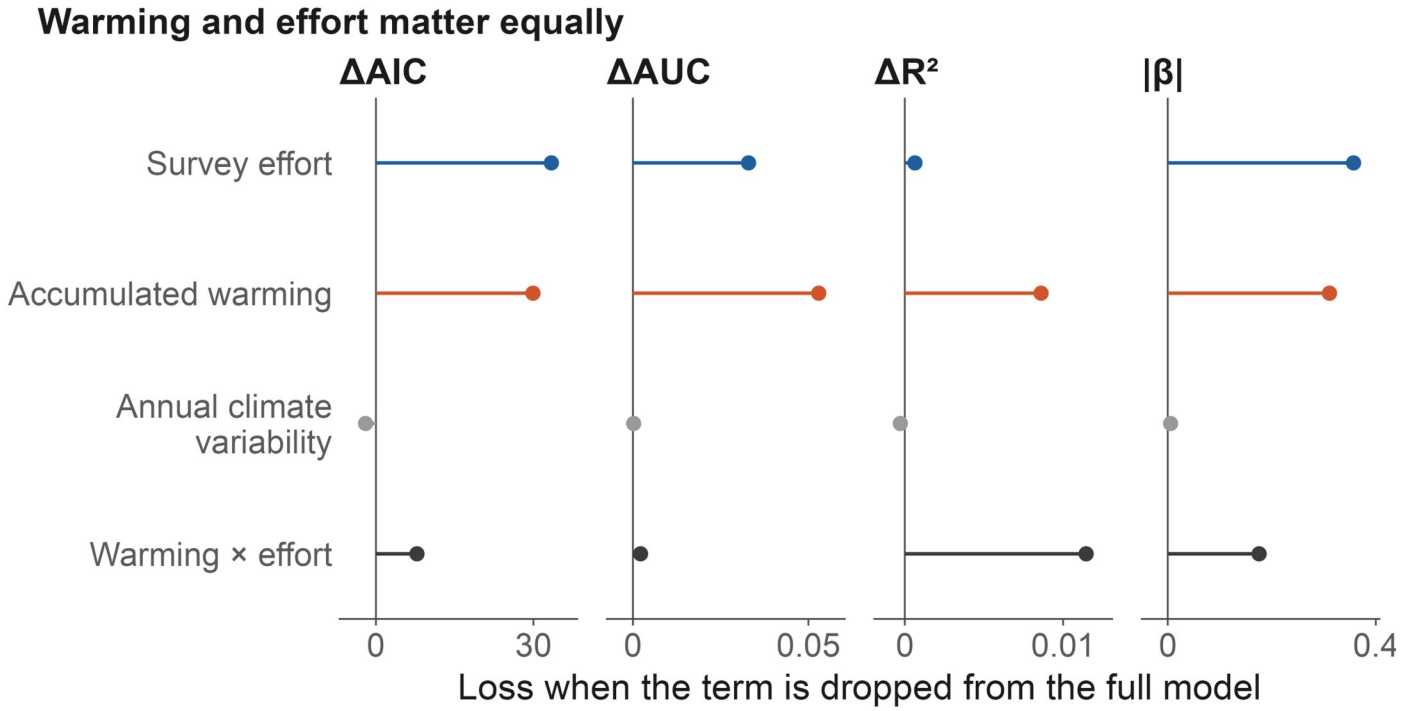


Fig. 1c Four criteria of importance

Warming and effort carry comparable weight under all four criteria

Cost of dropping a term

ΔAIC: warming 29.9, effort 33.4

ΔAUC: warming 0.053, effort 0.033

Conditional R^2 and |β| agree

Twelve specifications

SDM thresholds 50, 100, 200

4 effort proxies × 3 missing-data rules

Effort 1.25–1.55; warming 1.33–1.39

Design-preserving nulls

Warming null 1.16: increment ×1.18

Shuffled effort timing: 0.98

Province bootstrap: lower bounds > 1

◆ Finding 2

Survey pulses are the largest source of variation in the hazard

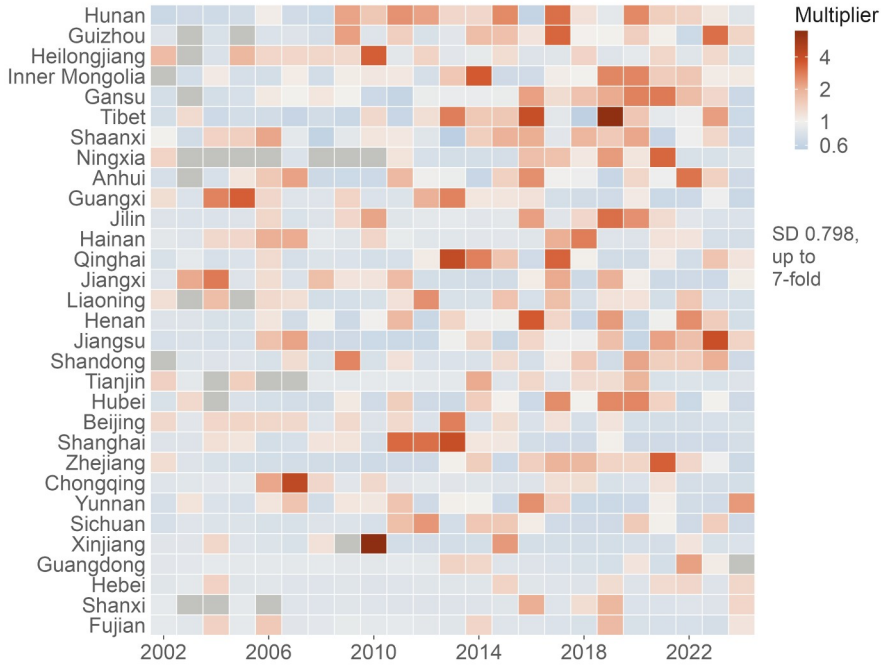


Fig. 5c Province × year multipliers

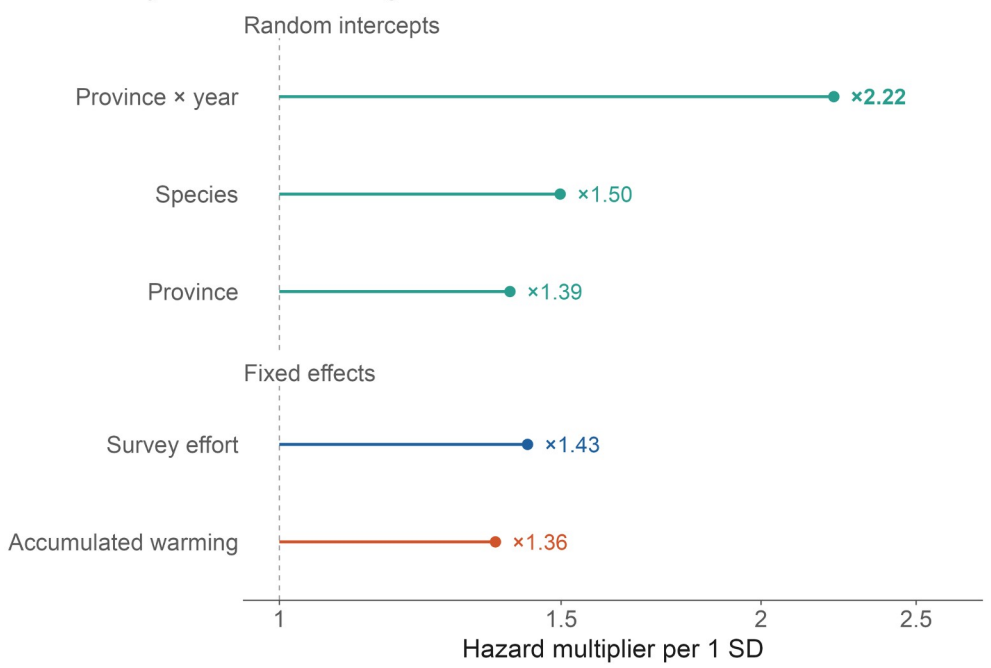


Fig. 4d Hazard multiplier per SD, all terms

Province × year
×2.22
per SD: survey pulses

Species
×1.50
intrinsic detectability

Province
×1.39
regional baseline

Conditional AUC
0.72 → 0.85
adding province × year

Finding 3

The return on effort is falling; the return on warming is not

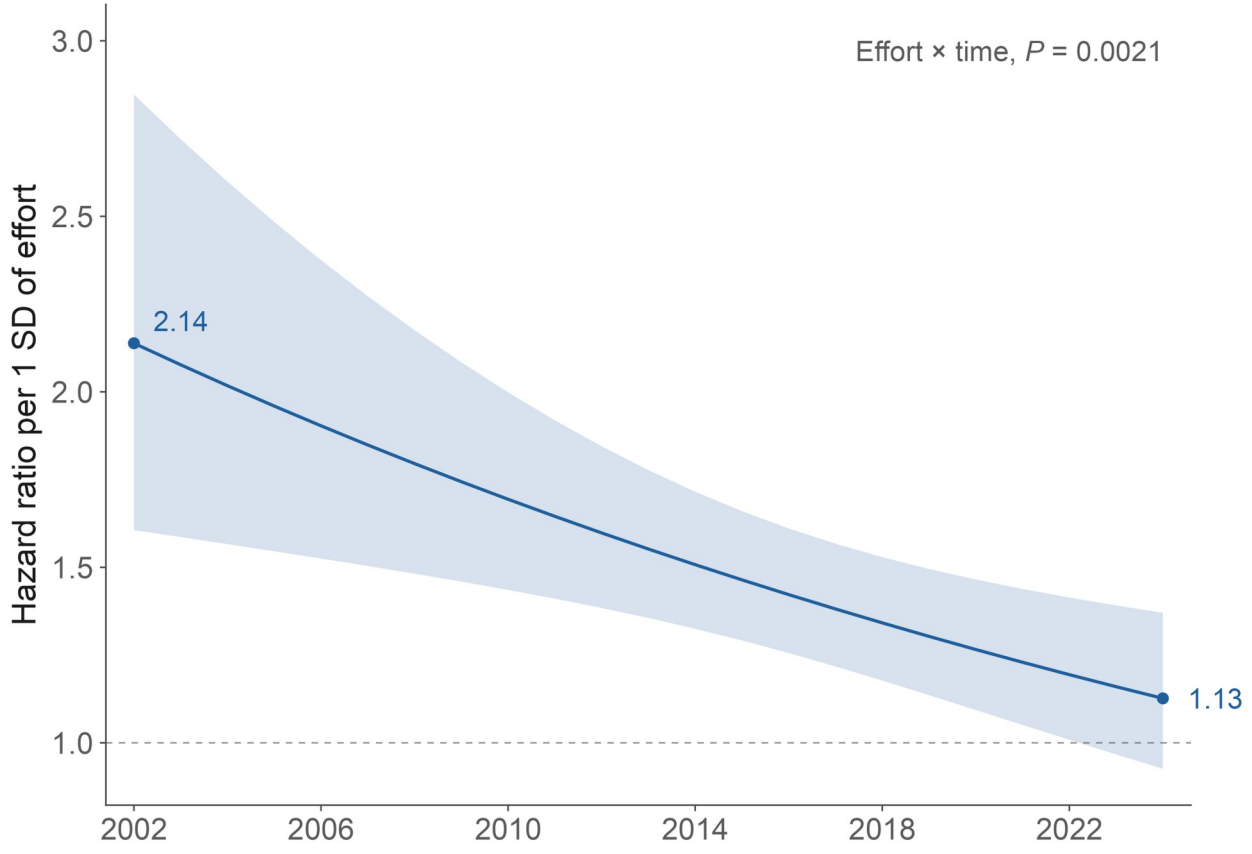


Fig. 5d Effort hazard ratio by year

Return on effort

2.14 → 1.13

HR per SD, 2002 → 2024; P = 0.0021

Return on warming

P = 1.00

warming × time: no trend; HR 1.37

A stock and a flow

Effort mines a finite stock of undocumented occurrences; warming keeps adding new candidates

Finding 4

The eastward bias is geometry; net of it, records lean slightly north

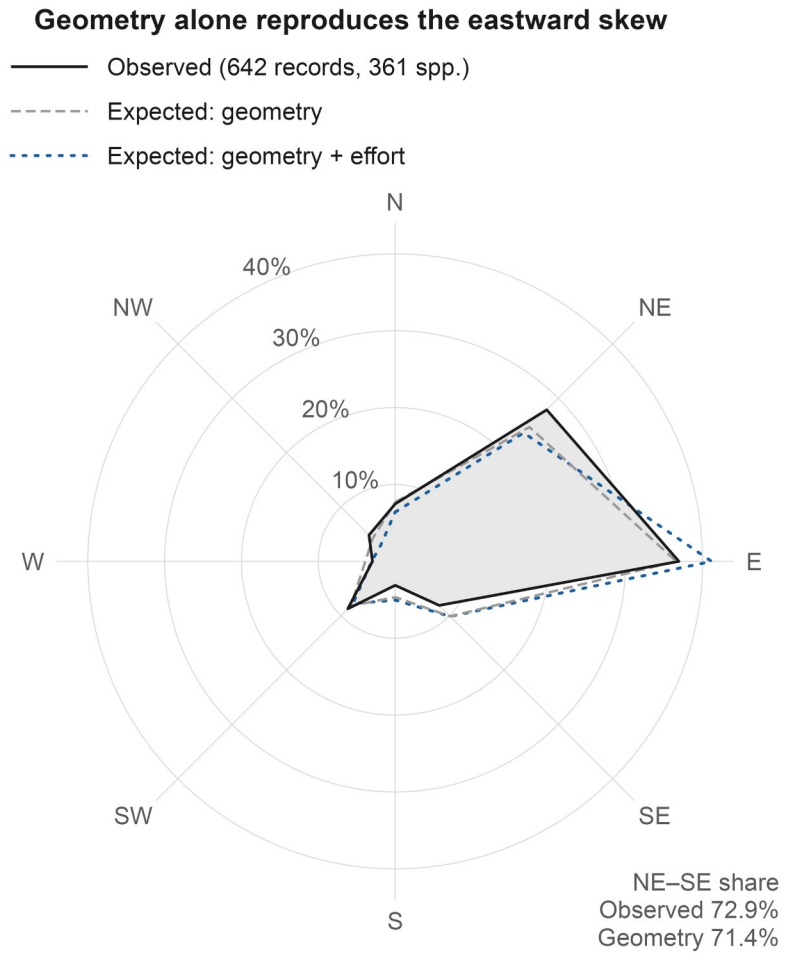


Fig. 6a Observed vs null sectors

Raw bearings, 642 records

East 36.9%, north-east 27.9%

China lies east of most ranges

Null 1: geometry only

East: 235 expected, 237 observed

No sector departs from the null

Null 2: geometry + birders

Northern sectors: 40.2% vs 32.8%

$\chi^2 = 26.8$, $P = 0.001$

A modest poleward lean

Application

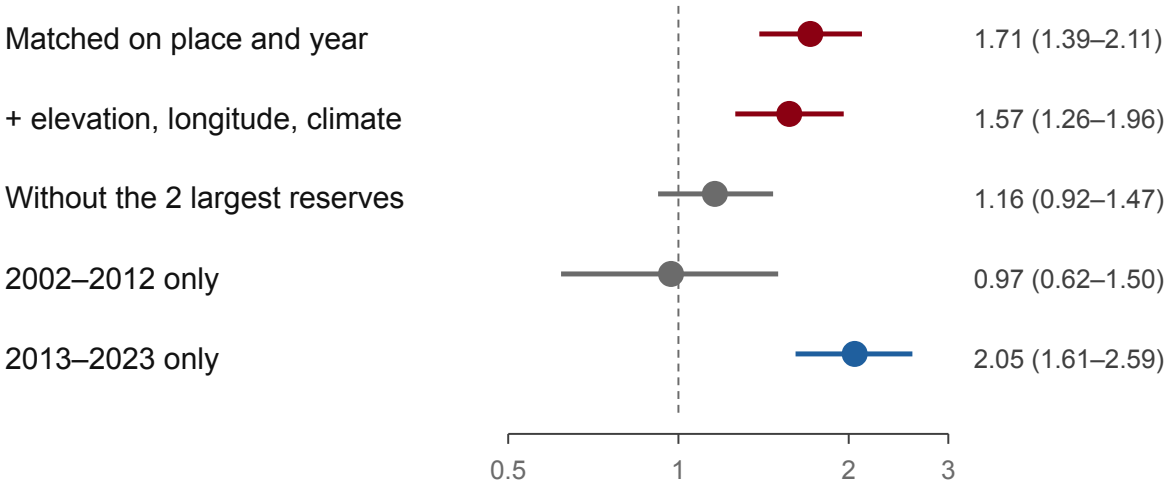
Records inside reserves track observation, not reserve performance

Share inside the 1,028 mapped reserves



Odds ratio against 200 matched controls

Controls from the same province and year, weighted by checklists; 95% CI



Few reserves carry the signal

172 events in 68 reserves

The top two hold 27.3% of them

Without them: OR 1.16

Only after birding boomed

2002–2012: OR 0.97

2013–2023: OR 2.05

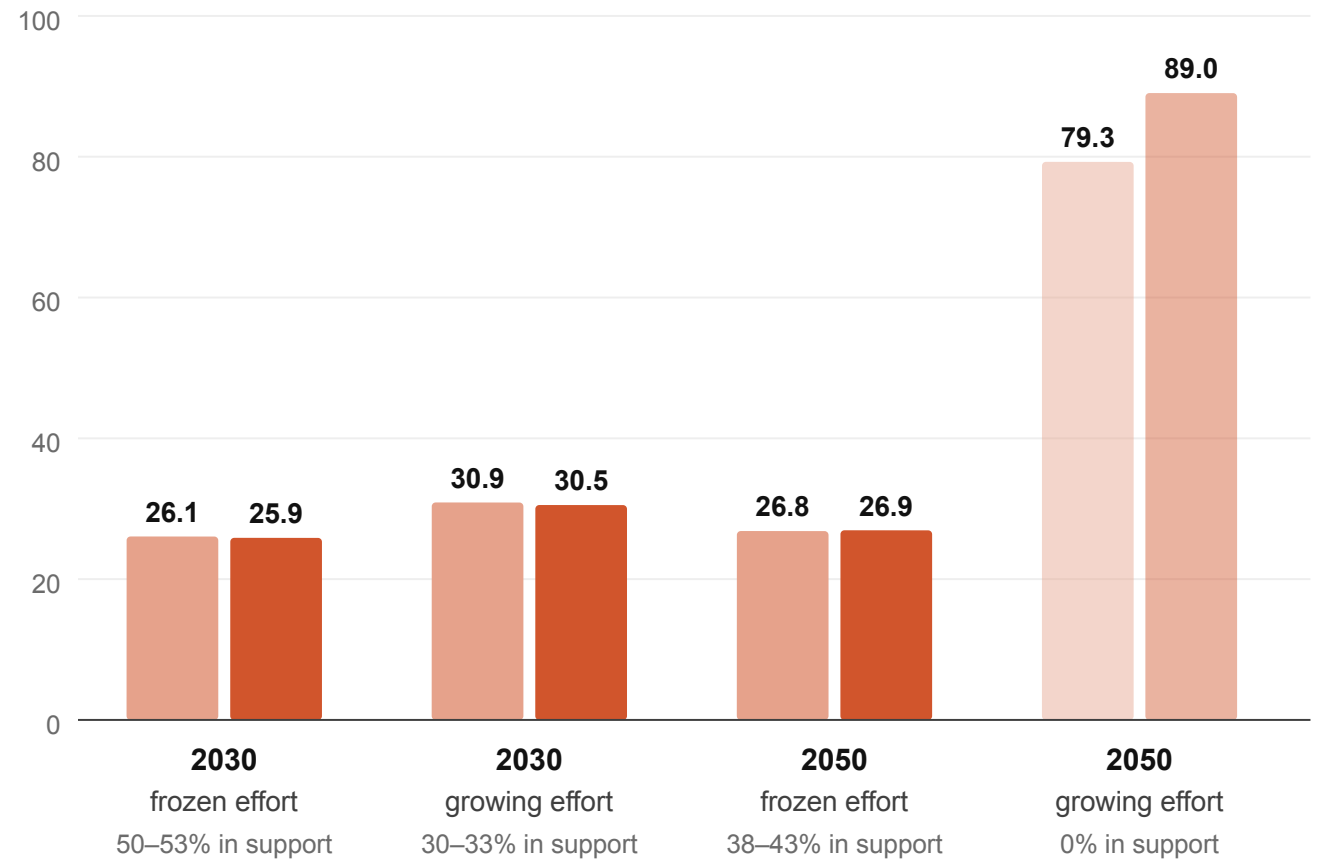
24% more species per visit inside

Record counts should not index reserve performance

Futures

Future record counts follow survey effort, not the emission scenario

Expected new records per year, China



Two climate × two effort scenarios

SSP2-4.5 or SSP5-8.5

Effort frozen, or still growing

What changes the count?

Climate scenario: barely anything

Growing effort: roughly triple

Most of that rise lies outside the fitted data (faded bars: none in support)

Futures

Data support, not ambition, sets the forecast horizon



Fig. 7b Hazard model vs machine learning

Support shrinks fast

Cells inside the fitted range:

2030: 46% → 2050: 6%

2080: none (SSP2-4.5)

Machine learning does not help

Agree inside the range (ρ = 0.56)

Diverge outside it

2030: usable
End of century:
illustration, not forecast

Limits

Four limits travel with these conclusions

Events are published records, not detections

We explain how records are documented, not range expansion itself.

Coordinates are precise to about 1 km at best

Reserve-overlap tests therefore use 0–5 km buffers as a sensitivity check.

Reserve polygons cover about a third of listed reserves

1,028 mapped reserves: no gap analysis, no 30×30 claims.

Projections leave the fitted support after 2030

Late-century maps are illustrations; the effort coefficient used is an upper bound.

Summary

Warming and survey effort leave signals of similar size

New records carry a climate signal mixed with an observation signal

01 Climate signal

Warming 1.36 vs effort 1.43 per SD: comparable, separable

02 Where few look

Warming × effort 0.84: under-surveyed regions hide the signal

03 Observation process

Province × year ×2.22; returns on effort 2.14 → 1.13

04 Futures

Emission scenario barely matters; most change is extrapolation

Implications

- ☐ Send survey effort to climatically exposed, under-surveyed regions
- ☐ Read counts of new records as observation-weighted evidence, not as range shifts
- ☐ Report where a model leaves its data support before mapping the future

Birds or birders? Both. Model how we look, and new records tell us how ranges change

Thank you

Questions welcome

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Open data and code

CBNR v2 (CC BY 4.0) · doi.org/10.5281/zenodo.20809949

github.com/dingchenchen6/bird-newrecord-discovery-dated

Climate: WorldClim 2.1 / CRU TS 4.09 · Ranges: BirdLife International
Effort: eBird/GBIF and China Bird Report · Base map: GS(2019)1822

We thank birdwatchers across China, and every author whose records made CBNR possible.

CBNR data

Backup

Dating to publication inflates effort and creates a phantom climate effect

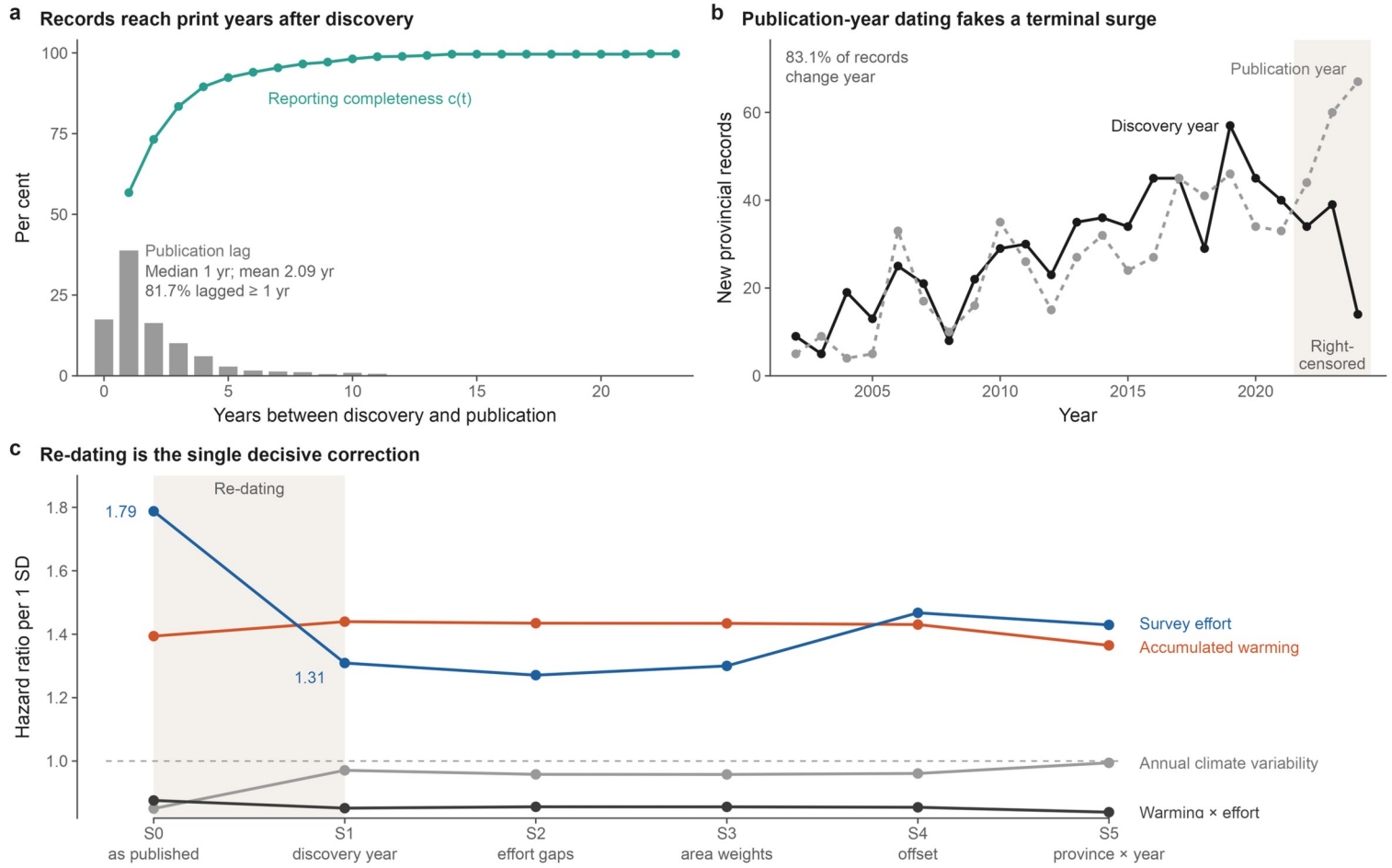


Fig. 2 Discovery vs publication dating

Backup

Specification robustness: climate indicator, window and effort proxy

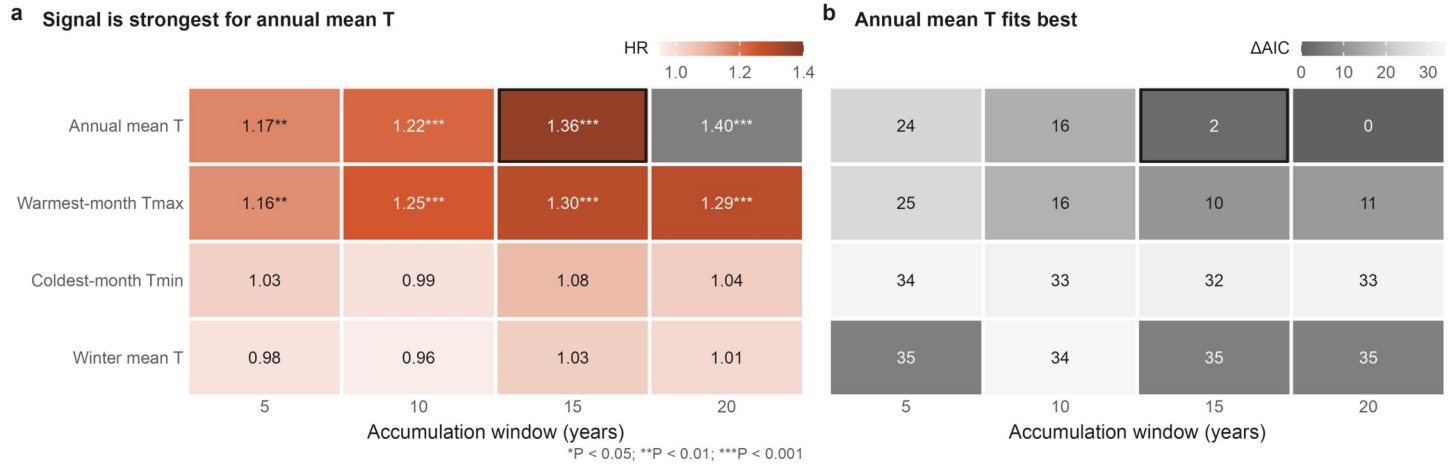


Fig. 3 Specification grid

Backup

Model anatomy: climate referenced to each species' range

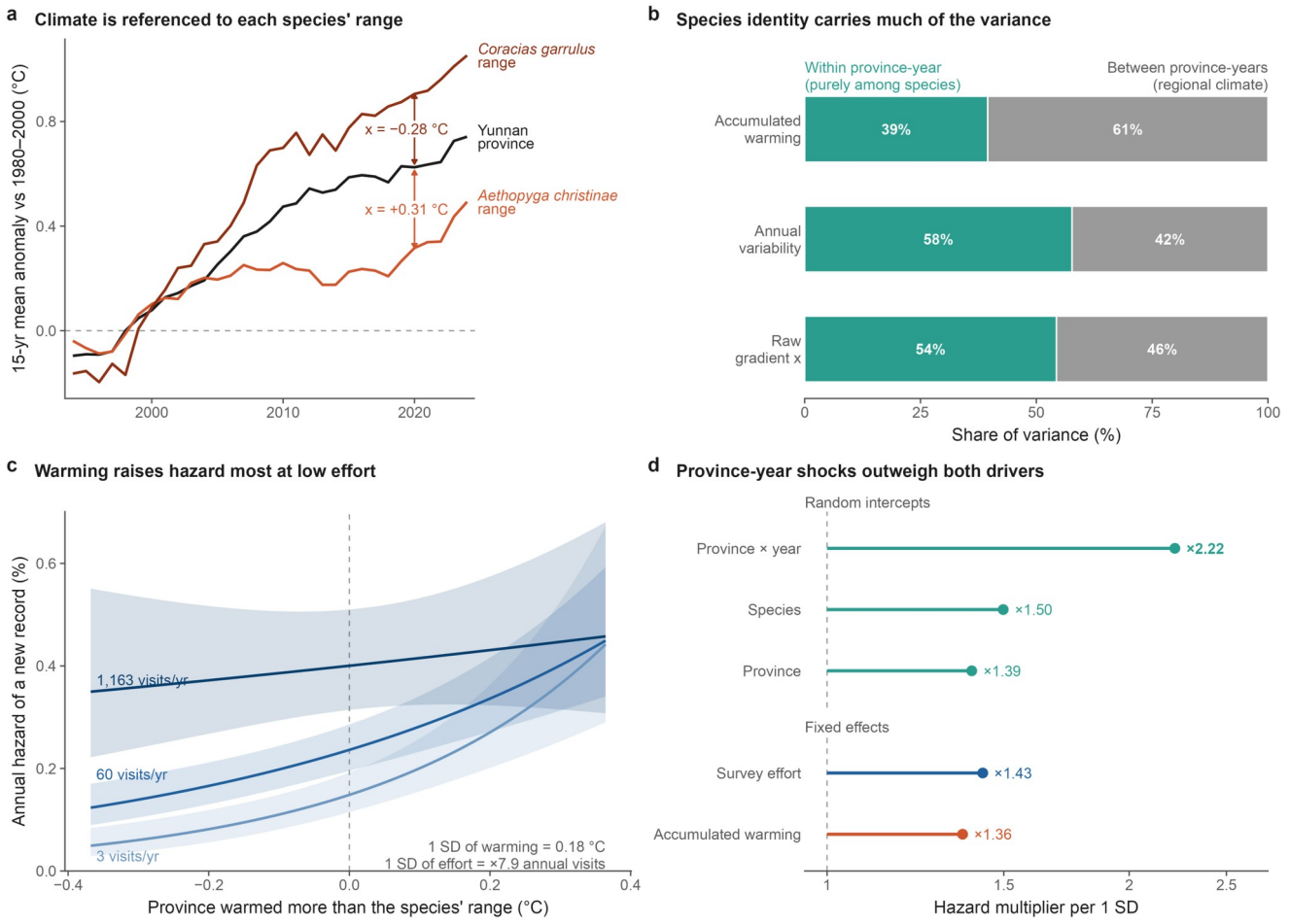


Fig. 4 Model anatomy

Backup

County-level surfaces: where to look first

